

CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents

Anonymous submission

Abstract

Long-horizon GUI agents can retain a complete interaction trace cheaply as textual action records, but expose only a few past events to the policy in high-fidelity pixels. We formulate this as *conditional fidelity restoration*: each event persists in summary-only form and is linked to an archived screenshot, while an active visual-context budget B limits how many events may be promoted to summary-plus-image form. Recent- B spends every slot on the latest events. CAUSALCACHÉ instead reallocates the same B promotions over the complete trace, evicting a recent image only when a distant event has higher conditional marginal utility. Its history-gated key/value (HGKV) adapter modifies only restored history-image tokens and is exactly bypassed with no history image. Matched-budget replacement groups and per-arm-anchored difference-in-differences supervision make uniform history amplification worth zero; a budget-aware selector then chooses which summarized events to restore. On desktop, the frozen policy shows no reliable target-over-recent preference ($+0.001$ nats/token, 95% CI $[-0.024, +0.025]$), whereas HGKV creates selectivity ($+0.010$ $[+0.006, +0.014]$) inside a pre-registered drift envelope. On OSWorld-Verified, exposing historical events in high fidelity is worth $+13$ – 14 pp over summary-only history, although same-budget allocations remain indistinguishable. Zero-shot on the 62-task construction-defined mobile memory-critical split, CAUSALCACHÉ improves over Recent-4 by $+8.6$ pp ($[+1.6, +16.1]$), remains neutral on matched controls, and yields a split-by-method interaction of $+10.4$ pp ($[+1.5, +19.6]$, $p=0.022$).

Introduction

GUI agents accumulate screenshots, actions, text arguments, coordinates, and interface changes over many decisions. A complete action trace persists cheaply as compressed text, but keeping every screenshot active is costly and distracting. Each past event therefore has two policy-visible fidelities: a summary-only record and, when needed, that same record with its archived screenshot reattached. The memory-control question is not only which events remain recorded, but which recorded events should be exposed again in high-fidelity pixels (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026).

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Our central claim is that the object of control is *event fidelity*, not event inclusion. Existing controllers retain recent events, retrieve visually similar screenshots, or learn salience from task supervision (Zeng et al. 2026; Shi et al. 2026; Liu et al. 2026). Yet the value of promoting a summarized event back to pixels depends on which images are already active: without recent images, a distant screenshot may lack local grounding; with a saturated recent window, another image may be redundant. Moreover, the action policy must consume the reattached pixels. Our experiments show that a strong frozen policy can respond to the *presence* of history images without being reliably sensitive to their *content*, leaving an otherwise better restoration policy with nothing to improve.

CAUSALCACHÉ accordingly treats GUI memory as *conditional high-fidelity restoration*. Every event remains in the complete low-fidelity action trace, and an archived screenshot is reattached only when that event is promoted into the active visual context; no pixels are reconstructed from text. Under a budget of B high-fidelity history images, Recent- B promotes the latest B events. CAUSALCACHÉ reallocates the same B promotions over the complete trace, with recent events as the default: a distant event enters only by displacing a recent image whose conditional marginal utility it exceeds. The matched- B comparison $Q(S^*) > Q(S_{\text{recent}})$ therefore isolates fidelity allocation from visual capacity. The realized number of non-recent promotions, $k = |S \setminus \text{Recent-}B|$, is a result statistic, not a preset parameter.

Learning to *use* reattached pixels must not corrupt the base agent. Full-layer adaptation simultaneously changes current-screen understanding, action syntax, and grounding. CAUSALCACHÉ instead trains a history-gated KV interface (HGKV) that touches only restored history-image tokens and is structurally bypassed at zero budget, with a difference-in-differences (DiD) objective in which every arm is anchored to its own frozen score: uniform amplification of any history cancels exactly, and only *selective* use of target-relevant content reduces the loss.

Our contributions are:

- a conditional fidelity-restoration formulation of GUI memory: all events persist as compressed action records, while Recent- B and CAUSALCACHÉ allocate the same B summary-plus-image promotions over that complete trace;
- a policy-preserving HGKV interface trained on matched-

budget replacement groups with per-arm-anchored DiD supervision, exact no-history parity, and pre-registered caps on recent-window and wrong-history drift, with a matched ungated control isolating the value of token gating;

- a budget-aware restoration selector over the complete summarized trace, trained on desktop data only and evaluated zero-shot on mobile and desktop benchmarks.

Related Work

Visual GUI agents and benchmarks. Screenshot-based GUI agents increasingly learn grounding and action prediction directly from pixels, including CogAgent, SeeClick, ShowUI, and GUI-Owl (Hong et al. 2024; Cheng et al. 2024; Lin et al. 2025; Xu et al. 2026). AndroidWorld provides dynamically instantiated mobile tasks with executable success checks (Rawles et al. 2025), and OSWorld evaluates agents in real desktop applications (Xie et al. 2024). GUI-Odyssey supplies cross-application mobile trajectories (Lu et al. 2025); our early mobile experiments on it motivated the conditional formulation used here. We use GUI-Owl-1.5-8B as the frozen cross-platform base policy (Xu et al. 2026), train on desktop trajectories from AgentNet-style successful demonstrations, and evaluate zero-shot on held-out desktop tasks and a cross-app mobile memory benchmark. **[Pending: Cite the MobileWorld benchmark report once its canonical reference is fixed.]**

Long-horizon GUI memory. GUI-Odyssey couples cross-app trajectories with a history resampler that attends to prior screenshots (Lu et al. 2025). MementoGUI learns multimodal memory writing, compression, and retrieval operators (Zeng et al. 2026). AndroTMem retrieves causally linked anchor states from interaction histories (Shi et al. 2026). These systems establish that long-horizon GUI control benefits from selective memory. Their primary control variable is which memory item is retained or retrieved. CAUSALCACHE instead keeps the event trace fixed in compressed form and controls the fidelity at which each event is exposed to the policy, conditioned on the visual context already active; it also trains the policy-side interface that consumes the restored pixels.

Interventional memory value. Recent work studies memory through controlled interventions or active state influence (Srivastava 2026; Liu et al. 2026). Our intervention is narrower and executable: every event always retains the same low-fidelity record, while a selected event is promoted by reattaching exactly one archived post-action image, holding the remaining prompt fixed. The image is retrieved from the archive, not generated from its textual record. The resulting score is a policy-specific behavioral surrogate rather than an environment-level causal effect. Coalition-conditioned marginals are related to cooperative-game attribution (Shapley 1953), but our goal is a deployable budgeted selector for a fixed GUI policy, not a generic attribution value.

Problem Formulation

At decision t , the agent receives instruction g , current observation x_t , and history

$$H_t = \{e_1, \dots, e_{t-1}\}, \quad e_j = (x_j, a_j, x_{j+1}), \quad (1)$$

where each event has a compact low-fidelity representation $e_j^{\text{low}} = \tilde{e}_j$ and an available high-fidelity representation $e_j^{\text{high}} = (\tilde{e}_j, v_j)$ formed by reattaching its archived post-action screenshot v_j .

Fixed active-fidelity budget. Every prompt contains the goal g , the *complete* textual action summaries $\{\tilde{e}_1, \dots, \tilde{e}_{t-1}\}$, and the current screenshot x_t . An active visual-context budget B limits how many summarized events may be promoted by reattaching their archived pixels; it is not a limit on cold archival storage. A restoration allocation is a set $S \subseteq \{1, \dots, t-1\}$ with $|S| \leq B$. Let $Q(S)$ denote the policy’s mean target-token log-likelihood of the successful next action when events in S are exposed in summary-plus-image form. The standard baseline is

$$S_{\text{recent}} = \text{Recent-}B, \quad (2)$$

the B most recent distinct events promoted to high fidelity. CAUSALCACHE targets

$$S^* = \arg \max_{\substack{S \subseteq \{1, \dots, t-1\} \\ |S| \leq B}} Q(S), \quad Q(S^*) > Q(S_{\text{recent}}), \quad (3)$$

so recency is one candidate fidelity allocation, not a separate modality. The replacement intensity

$$k = |S \setminus \text{Recent-}B| \quad (4)$$

is a measured statistic, and the reallocation curve

$$Q_B(k) = \max_{\substack{|S| \leq B \\ |S \setminus \text{Recent-}B| = k}} Q(S), \quad k = 0, \dots, B, \quad (5)$$

localizes where fidelity reallocation pays. All comparisons at a given B hold the image count, resolutions, summaries, and prompt structure fixed; only which summarized events are promoted differs. Cross- B comparisons are analysis, not the causal contrast. Given a partial allocation S , conditional marginals $\Delta_t(j \mid S) = Q(S \cup \{j\}) - Q(S)$ drive selection, and a replacement is justified only when the distant event’s restoration marginal exceeds that of the recent visual realization it displaces. Every formally reported set utility reruns the complete restored set; we never report a sum of singleton gains as the utility of a coalition.

CausalCache

Policy-Preserving History-Gated KV Adapter

Let h_l be the input to layer l ’s attention projections. In the last eight language-model layers, CAUSALCACHE adds rank-8, $\alpha=16$ low-rank residuals (Hu et al. 2022) only to key and value projections:

$$\begin{aligned} K_l &= W_l^K h_l + M_{\text{hist}} \Delta W_l^K h_l, \\ V_l &= W_l^V h_l + M_{\text{hist}} \Delta W_l^V h_l. \end{aligned} \quad (6)$$

M_{hist} is a token mask that is true only for image tokens reattached to promoted historical events. It excludes instruction tokens, event-summary text, the current observation image, and action tokens. The mask is constructed fail-closed from the multimodal token-type sequence and per-image patch geometry; any mismatch between declared images and encoded image blocks aborts the forward pass. When no historical event is promoted, the adapter context is absent and the projection hook returns the original module output, so $\pi_{\text{HG}}(\cdot \mid S=\emptyset) = \pi_0(\cdot \mid S=\emptyset)$ up to bitwise equality in our implementation.

Per- B Fidelity-Restoration Supervision

Training states are decision points of successful desktop trajectories in which the target action a_t^* recurs earlier in the same trajectory under full-action equivalence (type, arguments, and coordinates within tolerance); the screenshot preceding the earlier occurrence is the target-specific restoration candidate. For each state and each budget B we construct three matched-budget fidelity allocations with identical summaries, current screen, and target—every prompt carries exactly B history images:

- **Recent:** Recent- B , the B most recent distinct events promoted to summary-plus-image form;
- **Relevant restore:** Recent- $(B-1) + \{v_{j+}\}$, the oldest recent visual slot replaced by the target-specific archived screenshot (age $\geq B+2$, strictly older than the whole window);
- **Wrong restore:** the same slot replaced by an age-matched same-trajectory screenshot whose following action is *not* equivalent to a_t^* .

Positives do not require the base action to be wrong: both information addition and evidence amplification are admissible mechanisms.

Why single-slot replacement ($k=1$). Training teaches only the $k=1$ replacement interface, and this is an empirically grounded interface choice rather than a structural limit. On the demand side, multi-frame needs are rare even where archived pixels matter most: under the strictest *image-only* criterion—the required value appears in neither the goal, the current frame, the recent window, nor any completed step’s action text—mining 1,326 GUI-Odyssey trajectories yields 96 image-only decision points at $B=4$, of which only 2 require two distinct evidence frames simultaneously (3 of 239 at $B=1$); on the desktop corpus, 75% of eligible states admit exactly one recurrence witness. On the utility side, a matched-budget probe under the official multi-turn protocol shows the first replacement is worth its slot (+0.013, 95% CI [+0.005, +0.021] frozen log-likelihood margin) while paired increments of a second and third replacement are indistinguishable from zero or negative (+0.001 and −0.011, n.s.)—under a fixed budget each further distant promotion must evict the next-most-recent visual realization, and its measured value no longer covers that price. Larger- k allocations nonetheless remain expressible at deployment: the selector composes exactly- B allocations with exact re-scoring, defaulting unclaimed slots to recent events, so k stays a result

statistic. Because supervision is annotation-free—labels are teacher-forced counterfactual scores of the current policy on its own trajectories—a domain that does exhibit multi-frame demand (detectable by the same probe) requires only regenerating the corpus with $k \geq 2$ restoration arms and re-running the identical trainer: a data-mix change, not a method change.

Let ℓ denote mean target log-likelihood, superscripted by adapter state (active or frozen bypass). Each arm is anchored to *its own* frozen score:

$$\begin{aligned} A_s &= \ell_{\text{relevant}}^{\text{on}} - \ell_{\text{relevant}}^{\text{off}}, & A_r &= \ell_{\text{recent}}^{\text{on}} - \ell_{\text{recent}}^{\text{off}}, \\ A_n &= \ell_{\text{wrong}}^{\text{on}} - \ell_{\text{wrong}}^{\text{off}}, \end{aligned} \tag{7}$$

and the DiD objective is

$$\begin{aligned} \mathcal{L} &= [m - (A_s - A_r)]_+ + [m - A_s]_+ \\ &\quad + [m - (A_s - A_n)]_+ \\ &\quad + \lambda_{\text{cap}}([|A_r| - \epsilon]_+ + [|A_n| - \epsilon]_+) + \lambda \|\Delta W\|_2^2, \end{aligned} \tag{8}$$

with $m=0.01$, $\epsilon=0.02$, $\lambda_{\text{cap}}=2$, $\lambda=10^{-4}$, and no action cross-entropy. On the identity adapter every increment is zero, so $A_s - A_r$ is exactly zero: the frozen policy’s own preference for target-relevant restorations cannot masquerade as adapter skill, and uniformly amplifying all restored pixels buys nothing. The dead-zone caps bound recent-window and wrong-history drift with gradients on the same scale as the selection hinges. Losses and evaluation are stratified by B ; a pooled mean could hide a method that helps at $B=4$ but fails at $B=1$.

Success-Anchored Utility and Selector Supervision

After the history interface is frozen, selector labels are true policy utilities: for each state, singleton utilities for the *entire* summarized event pool (recent events included)—an exhaustive $B=1$ oracle—the matched $Q(\text{Recent-}B)$ anchors, and, for larger budgets, set utilities along beam-3 teacher paths over a singleton-shortlisted pool. Every set is fully re-scored as a complete prompt through the frozen HGKV policy; set utility is never approximated by summing singleton utilities.

Budget-Aware Restoration Selector

The selector is a single set-conditional marginal scorer $\hat{\Delta}_t(j \mid S)$ trained on the true marginals implied by the label table (singleton edges give $S=\emptyset$; each beam edge gives its parent set), with a joint regression and within-state ranking loss. Its input encodes, explicitly, candidate age and recency-window membership, target-recurrence witness signals with coordinate distance, action family, frame-deduplication multiplicity, trajectory shape, and set context (selected-set size, overlap and age gaps between the candidate and the already-selected events).

Deployment composes exactly- B fidelity allocations by beam search initialized at Recent- B : every summarized event is a candidate, and a distant event is promoted only by displacing the recent visual realization whose predicted marginal it beats, realizing Eq. 3 as a relative comparison with no learned stopping threshold—declining every replacement recovers Recent- B exactly. Selector labels cover

$B \in \{1, 2, 4\}$; the realized replacement intensity k (Eq. 4) is reported per budget, never preset.

Reference actions and the two deployment modes. Four of the selector’s input features are *witness* features: they test whether the action taken right after some archived event matches a reference action. At training time the reference is the gold teacher-forced target—an offline oracle that upper-bounds what any online reference can provide but does not exist at deployment. We therefore instantiate two online modes that share every trained weight and differ only in the reference. **CausalCache-P** (default) uses the current policy’s tentative action: the policy first acts on the Recent- B prompt, the tentative action serves as a platform-calibrated reference, and the policy re-decides on the re-allocated prompt when a recent visual realization is displaced. **CausalCache-LA** replaces the proposal with *the action just executed* at step $t-1$, eliminating the second policy call: memory is selected before the single generation, adding only the selector’s own cost (< 10 ms). On held-out desktop decision groups the two modes are statistically indistinguishable (LA $+0.015$ [$+0.006, +0.025$], gold-reference $+0.011$ [$+0.001, +0.021$] over Recent-4), but LA fails to transfer to the unseen mobile benchmark, where P is significantly better (Section); we therefore treat LA as an efficiency variant rather than the primary method, and the result is consistent with the reference action being one of the bottlenecks for cross-platform transfer.

A ladder of weaker references isolates what the action signal buys. *No reference* (witness features zeroed) keeps $B=2$ significant ($+0.016$ [$+0.003, +0.030$]) but loses $B=4$ ($+0.009$ [$-0.002, +0.020$]) while more than tripling full-replacement allocations ($k=4$: $11\% \rightarrow 34\%$)—without any action signal the selector cannot tell which summarized event merits visual restoration and over-replaces. *Instruction-only intent* (eight frozen-embedding cosine features between the instruction, each candidate’s action line, and recent behavior) repairs the over-replacement pathology ($k=4$ back to 12%) yet stays non-significant at $B=4$ ($+0.005$ [$-0.007, +0.016$]). Action-conditioned references thus dominate instruction-only and reference-free variants; the relative order of the action-conditioned references themselves depends on platform and protocol, which is precisely why the deployed default re-grounds the reference in the current policy’s own proposal.

Experimental Setup

Training Data and Model Selection

Policy adaptation and selector training use only desktop data: successful AgentNet/OpenCUA Ubuntu trajectories (5,000 screened, 2,293 successful, 6,003 decision points) yield fixed-budget replacement groups at $B=1/2/4$ of 969/764/470 (2,203 in total; 1,774 training units and 211 development groups, trajectory-disjoint; click 52–56%, plus key/type/drag/scroll), complemented by a high-precision OSWorld witness seed mined from official successful trajectories. A further 160 groups at $B=8$ are constructed but deliberately excluded from training, so $B=8$ evaluates out-of-training budget extrapolation. Checkpoints and thresholds

B	Recent baseline	Replacement arms	CAUSALCACHE
1	Recent-1	prev. frame $\rightarrow$ archived	Selected-1
2	Recent-2	oldest slot $\rightarrow$ archived	Selected- ≤ 2
4	Recent-4	oldest slot $\rightarrow$ archived	Selected- ≤ 4
8	Recent-8	(built, not trained)	Selected- ≤ 8

Table 1: The fixed-budget fidelity-restoration design. Training arms demote the oldest recent event to summary-only form and promote one archived event to summary-plus-image form ($k=1$). The deployed selector allocates all B active visual slots over the complete summarized trace, with recent events as the default, and reports the realized k . All contrasts at a given B are matched in image count; cross- B comparisons are analysis only.

are selected only on the desktop development split under pre-registered gates: positive DiD selection with a positive bootstrap lower bound, and drift caps $|A_r|, |A_n| < \epsilon$. No mobile benchmark data touches training or selection.

The Fixed-Budget Fidelity-Restoration Design

Table 1 defines the core comparison. Each row fixes the budget B ; every compared allocation carries exactly B history images with identical resolutions, summaries, and prompt structure, so the contrast isolates *which summarized events are promoted to high fidelity*. $B=4$ is the primary deployment-like setting.

Policies and Fidelity-Allocation Baselines

The frozen policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026). We compare four policy rows under identical data, steps, and cadence:

- **Frozen**: no trainable adapter;
- **Ungated KV**: the same last-eight layers, K/V targets, rank, and alpha as HGKV but active on all tokens;
- **HGKV (ours)**: token-gated, structurally bypassed at $B=0$.

Fidelity-allocation baselines are Recent- B , Random, OCR/RGB similarity, and the learned budget-aware restoration selector (exact- B , recent fallback); a beam oracle over the candidate pool bounds headroom. Every method receives the same summaries and the same active image budget.

Zero-Shot Benchmarks and Metrics

The policy interface and selector, trained on desktop data only, are evaluated (i) on held-out desktop decision groups (offline, per- B), (ii) zero-shot on an uncontaminated OS-World task roster (Xie et al. 2024) with official evaluator scores (tasks whose trajectories seeded the witness corpus are excluded from zero-shot claims and reported separately as in-domain diagnostics), and (iii) zero-shot on a cross-app mobile memory benchmark of 117 runnable tasks (62 cross-app memory candidates, 55 single-app controls), which contributes no data to any training or selection decision. **[Pending: Fix the MobileWorld citation and the final roster**

hashes.] Offline metrics: mean target log-likelihood margins, full-action equivalence (type, arguments, coordinates at $\tau=25$ of a $[0, 999]$ grid; $\tau \in \{5, 10\}$ as sensitivity), repair and amplification rates, wrong-history harm, parser validity, and active history-image count. Closed-loop metrics: paired task success with cluster bootstrap, step counts, and wall time.

Results

Selectivity Is Not Inherited from the Frozen Policy

On the desktop development split at $B=1$ (94 groups, 77 trajectories), the choice of recent baseline matters. Against a *redundant* baseline that promotes a byte-identical copy of the current screen (the deployment recent-selector’s behavior), the frozen policy prefers the target-relevant archived screenshot by $+0.0333$ nats/token (95% CI $[+0.0118, +0.0567]$). Against the *informative* true previous frame, however, its preference vanishes: $+0.0013$ $[-0.0242, +0.0253]$, not significant (94/94 probes have a previous frame that differs from the current screen). To the frozen policy, promoting a task-relevant distant event and retaining the recent event it would displace are worth about the same; profitable fidelity reallocation must be learned. Under the official multi-turn protocol the frozen replacement effect is $+0.079/ +0.026/ -0.000/ -0.031$ at $B=1/2/4/8$: whatever native preference exists decays with the window and reverses, while the learned interface keeps a positive selection effect across the whole budget axis.

Selective Use Within a Drift Envelope

Table 2 reports the pre-registered desktop DiD gate at $B=1$. All three adapters are trained for 150 matched steps; checkpoints are selected by the frozen rule (all gates passed, maximal DiD selection, earliest tie-break). HGKV passes every gate from step 50 onward: its selected checkpoint improves DiD selection to $+0.0111$ $[+0.0074, +0.0150]$ against the redundant-duplicate recent control, and a re-scoring probe confirms $+0.0095$ $[+0.0056, +0.0138]$ (60/94 groups) against the informative true-previous-frame control, while holding recent-only and wrong-history drift well inside $\epsilon=0.02$; its no-history path is bitwise identical to the frozen policy under randomized nonzero adapter weights. The layer- and rank-matched ungated control also learns selectivity ($+0.0076$), confirming that the last-eight K/V position carries signal, but token gating adds a paired per-group advantage of $+0.0035$ (95% CI $[+0.0007, +0.0062]$, 56/94 groups) while holding drift further from the caps. Gating is what buys selectivity *inside* the envelope.

Fidelity Reallocation and Zero-Shot Transfer

Per- B DiD gates all pass for HGKV at every budget, including the untrained $B=8$ extension: $\text{did}_{\text{select}} = +0.0224$ $[+0.0162, +0.0294]$, $+0.0164$ $[+0.0110, +0.0221]$, $+0.0109$ $[+0.0070, +0.0154]$, and $+0.0063$ $[+0.0018, +0.0115]$ for $B=1/2/4/8$, with recent-window and wrong-history drift within the pre-registered caps throughout. A layer- and rank-matched ungated KV control also passes its gates but attains uniformly smaller

Adapter (selected step)	DiD select	$ A_r $	Wrong drift	Gates
HGKV (s150)	+0.0111	0.0087	0.0085	pass
Ungated KV (s125)	+0.0076	0.0090	0.0063	pass

Table 2: Desktop DiD gate on 94 development groups ($B=1$; 2,000 episode-clustered bootstrap replicates). DiD select = $(A_s - A_r)$; caps require $|A_r|, |A_n| < 0.02$. HGKV’s DiD CI is $[+0.0074, +0.0150]$; HGKV B0 parity is bitwise exact.

selection effects ($+0.0177/ +0.0139/ +0.0075/ +0.0047$) and drifts toward the cap at $B=8$ ($|A_r|=0.016$ vs. 0.009 for HGKV): token gating adds selective gain and containment beyond matched capacity. The frozen replacement effect decays from $+0.079$ at $B=1$ to -0.031 at $B=8$, while the learned end-to-end selector beats Recent- B at matched image counts by $+0.030$ $[+0.019, +0.041]$ ($B=2$) and $+0.017$ $[+0.007, +0.028]$ ($B=4$), keeping the full recent window in 19% of $B=4$ states.

Mobile closed loop (zero-shot). The 117-task mobile benchmark — which contributes no training or selection signal — is our primary test of *allocation*, because its roster is constructed with a memory-critical split. The full-roster aggregates are secondary: CAUSALCACHE-P attains the highest overall success rate, 33.9% (three runs: 39/42/38 of 117) against 30.2% for HGKV+Recent-4 (three runs: 34/37/35; paired $+3.7\text{pp}$, 95% CI $[-0.9, +8.6]$, per-round $+4.3/ +4.3/ +2.6\text{pp}$) and 28.2% for the frozen summary-only baseline (three runs: 33/32/34; $+5.7\text{pp}$, $[-0.6, +12.0]$). The decomposition is clean: frozen Recent-4 (29.9%), HGKV+Recent-4 (30.3%), and the summary-only baseline are statistically indistinguishable — the behavioral counterpart of the drift cap — so the gain is carried by *which summarized events are promoted*, not by the adapter or by recency alone. A full recent-dose sweep of the frozen policy makes the last point explicit: across $B=0/1/2/3/4$ the success rate is 28.2/26.5/26.5/29.1/29.9%, non-monotone with every paired contrast against $B=0$ crossing zero — on this benchmark, promoting additional recent events alone buys nothing, so fidelity *allocation* rather than active visual budget *size* is the operative variable. The primary closed-loop result is the memory-critical split, defined at benchmark construction: 62 cross-application tasks whose target information surfaces in an application visited *earlier* in the episode — after the app switch, the Recent- B window no longer contains that evidence by construction — and 55 matched single-application controls with no such dependency. On the memory-critical stratum the same-budget contrast is decisive: CAUSALCACHE-P 28.0% vs. Recent-4 19.4%, $+8.6\text{pp}$ (95% CI $[+1.6, +16.1]$); on the controls the method is correctly inert (40.6% vs. 42.4%, -1.8pp $[-6.7, +3.6]$); the split-by-method interaction is $+10.4\text{pp}$ (95% CI $[+1.5, +19.6]$, two-sided $p=0.022$). The value of reallocation is conditional on distant visual dependency: the modest full-roster mean is composition, not absence of effect, since 47% of the roster is control tasks on which no high-fidelity restoration method should help.

We further define *memory-demanding long-horizon tasks*

as tasks that the frozen summary-only policy cannot complete within the benchmark’s official interaction budget, and report $\Delta_{\text{long}} = P(\text{success} \mid \text{CAUSALCACHE, long}) - P(\text{success} \mid \text{baseline, long})$ on that stratum. On the 95 such mobile tasks (50-step budget), CAUSALCACHE-P attains 30.9% with $\Delta_{\text{long}} = +7.0\text{pp}$ over the summary-only baseline (95% CI $[+0.4, +14.0]$) and $+3.5\text{pp}$ over frozen Recent-4 at the same budget ($[-3.2, +10.2]$); on the 22 short tasks the baseline completes, memory reallocation adds nothing (47.0% base rate). The allocation benefit concentrates on construction-defined memory-critical tasks, not merely long tasks. The efficiency variant CAUSALCACHE-LA, unchanged in weights, fails to transfer (29.1%; -4.8pp vs. P, 95% CI $[-10.5, +0.6]$), and its failure concentrates on single-app control tasks where the last-action reference trivially recurs: under platform shift the proposal reference is what carries the selection signal.

Desktop closed loop. On OSWorld-Verified (361 tasks, official no-GDrive roster, docker provider, the official default budget of 15 steps per task applied uniformly to all arms), promoting historical events to high fidelity improves substantially over summary-only memory, while alternative same-budget allocations remain indistinguishable: CAUSALCACHE-LA 34.1%, CAUSALCACHE-P 32.7%, HGKV+Recent-4 33.0%, frozen Recent-4 32.7% (all pairwise differences non-significant), each $+13\text{--}14\text{pp}$ over the frozen summary-only baseline (19.9%; paired 95% CIs excluding zero, $p < 10^{-4}$). OSWorld thus supports the high-fidelity *availability* claim; the conditional fidelity-*allocation* claim is tested by the mobile memory-critical split, and we do not claim that reallocation helps on every long task. The adapter is again behaviorally neutral on matched allocations (HGKV+Recent-4 vs. frozen Recent-4: $+0.3\text{pp}$, $p=0.96$), mirroring the mobile finding and giving the drift cap its behavioral certificate on both platforms. Per-task cost instrumentation of the P arm gives the deployment overhead directly: median 8 extra policy calls and 20.6 model-seconds per task against a median task wall time of 308s — an end-to-end overhead of 7% at the median (10% at P90, and 2.7% on successful tasks), even though pass-2 fires on 82% of steps and adds 66% to policy compute in isolation.

Ablations and Analyses

Ablations follow the fixed-budget fidelity factorization. Policy-side, the per- B stratified DiD gates for HGKV and ungated KV and the $B=8$ extrapolation appear in the Results; the frozen recent-dose sweep (varying window size without replacement) appears in the mobile closed-loop analysis. Selector-side, a chooser ladder brackets the learned selector at matched budgets (selector–Recent- B , episode-clustered CIs): the deployed LA selector scores $+0.025 / +0.015$ at $B=2/4$; *random* exact- B restoration is indistinguishable from Recent- B ($+0.006 [-0.006, +0.017] / +0.008 [-0.001, +0.018]$), as is *RGB-similarity* retrieval ($+0.006 / +0.007$, both crossing zero)—visual similarity to the current screen is not utility; *inverting* the learned scorer (bottom- k) is significantly harmful ($-0.016 [-0.029, -0.003] / -0.009 [-0.018, -0.000]$), a specificity control showing

the scorer ranks real signal symmetrically; and retraining without the recency-membership features degrades the selector to the noise band ($+0.012 [-0.000, +0.025] / +0.008 [-0.001, +0.017]$), so budget-context features carry substantial weight. The witness-reference ladder (gold/proposal, last-action, instruction-only, none) appears in the deployment section.

Limitations and Claim Boundary

Our claim is about conditional fidelity restoration under an active context budget:

Every completed event remains available as a compressed action record. Recent- B promotes the latest B events to summary-plus-image form. CAUSALCACHE instead allocates the same B high-fidelity promotions over the complete summarized trace, reattaching archived screenshots rather than generating pixels from text. Its benefit therefore comes from reallocating an active visual-context budget, not from adding images or expanding archival storage.

The claim is *not* that archived screenshots are usually better than recent ones; it is that for some decisions, the conditional value of promoting a specific summarized event exceeds that of the recent visual realization it displaces, and CAUSALCACHE learns to recognize those states and events. The teacher-forced likelihood surrogate must ultimately be judged by closed-loop success. “Causal” refers only to controlled fidelity interventions that reattach archived pixels while holding the remaining prompt fixed, not environment-level structural causality. The training-time $B=1$ recent arm followed the deployment selector, whose top slot duplicates the current screen, so we separately re-scored the selected checkpoint against the true previous frame and report both controls; the v3 corpus freezes recent slots as distinct frames and discloses the deployment duplication separately. Finally, we cannot guarantee that the pretrained base policy never saw related data.

Conclusion

CAUSALCACHE treats long-horizon GUI memory as conditional high-fidelity restoration. The complete event trace remains available in compressed text, while a fixed active-context budget determines which summarized events regain their archived pixels; recency is the default fidelity allocation, not a privileged memory modality. A token-gated KV interface learns to consume those restored pixels inside a pre-registered drift envelope, and a budget-aware selector learns when promoting a distant event is worth demoting a recent one. All learning happens on desktop data; mobile benchmarks remain untouched for zero-shot evaluation. In one sentence: CAUSALCACHE learns which summarized GUI events deserve to be seen again.

Fixed-budget fidelity allocation (offline desktop heldout; primary setting $B = 4$)

Policy	Allocation	$B=1$	$B=2$	$B=4$	$B=8^\dagger$	Realized k	Notes
Frozen	Recent- B	0	0	0	0	0	reference
Frozen	oldest slot $\rightarrow$ archived	+0.079	+0.026	-.000	-.031	1	
HGKV	Recent- B	+0.002	+0.005	+0.007	+0.009	0	drift-capped
HGKV	oldest slot $\rightarrow$ archived	+0.103	+0.047	+0.018	-.016	1	gate: Tab. 2
HGKV	selector, LA reference (deployed single-pass)	+0.039	+0.025	+0.015	-.005	measured	71% displace ≥ 1 at $B=4$
HGKV	selector, gold reference (offline upper bd.)	+0.044	+0.025	+0.011	-.002	measured	teacher-forced oracle

Table 3: Skeleton of the fixed-budget main table. Each cell is a matched-count contrast on the desktop heldout split; every allocation promotes exactly B events to summary-plus-image form, and the learned selector may keep every promotion on the recent window ($k=0$); $^\dagger B=8$ is excluded from training and reports budget extrapolation. Cross- B comparisons are analysis only.

Method	OSWorld-Verified (361)	MobileWorld (117)	Extra policy calls
Frozen, summary-only ($B=0$)	19.9	28.2 (3 runs)	0
Frozen + Recent-4	32.7	29.9	0
HGKV + Recent-4	33.0	30.2 (3 runs)	0
CAUSALCACHE-LA (single-pass)	34.1	29.1	0
CAUSALCACHE-P (default)	32.7	33.9 (3 runs)	conditional

Table 4: Closed-loop success rates (%). OSWorld-Verified: official no-GDrive roster, docker provider, official 15-step budget, one run per arm; all high-fidelity restoration arms beat summary-only $B=0$ by +13–14pp (paired 95% CIs excluding zero) and are pairwise indistinguishable. MobileWorld (zero-shot): three-run means; CAUSALCACHE-P over HGKV+Recent-4 is +3.7pp (95% CI $[-0.9, +8.6]$; per-round +4.3/ +4.3/ +2.6pp) and +5.7pp over $B=0$ (95% CI $[-0.6, +12.0]$). CAUSALCACHE-P’s second call fires on 82% of steps but adds only 7% median end-to-end task time. Per-round results and full paired statistics are in the appendix.

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